Assessment (Week 3)

Description

Module/Week3 – Assignment 2

Topic: Structured Query Language; Join Operations; Beyond Simple SELECT Queries

Student's Full Name: Vishal Guddeti

Course Title: INFO 531 - Data Warehousing and Analytics in the Cloud

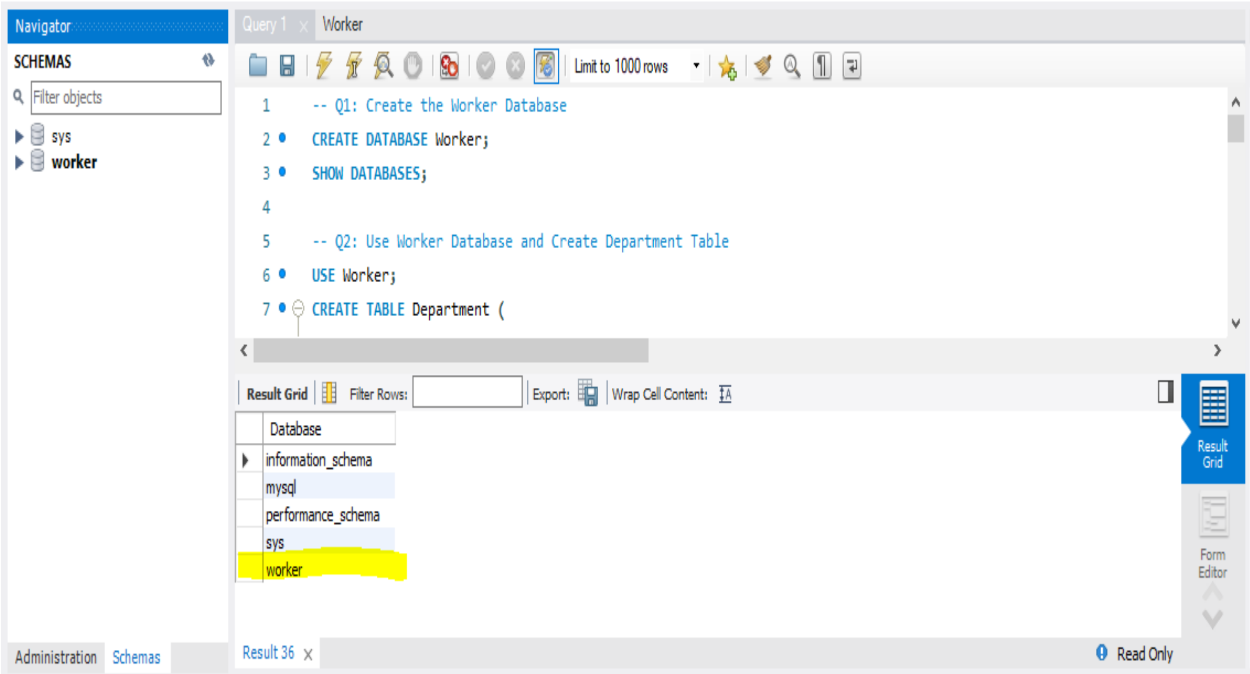
Term name and year: Fall 2023

Submission Week: , Week 3- Assignment 2

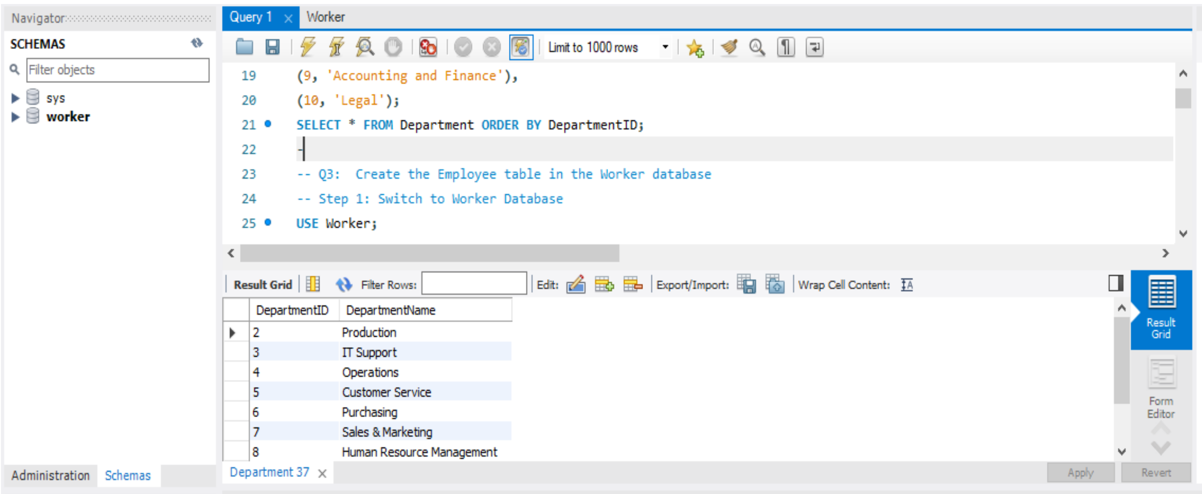
Instructor's Name: Dr.Nayem Rahman

Date of Submission:31-07-2024

Q1. Show the screenshot of a successful installation of MySQL Software and MySQL Workbench with the latest version on your machine. Show the screenshot of the database "Worker"created.

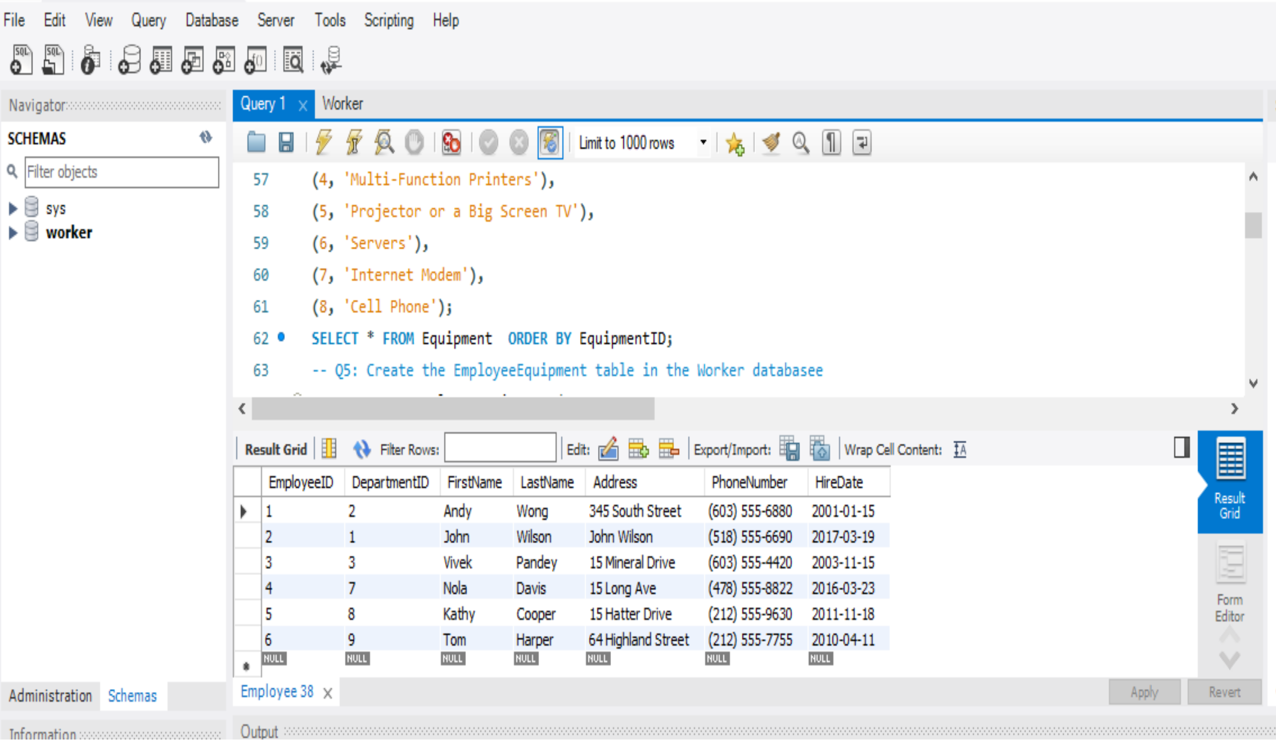


Q2. . Create the Department table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Department table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).

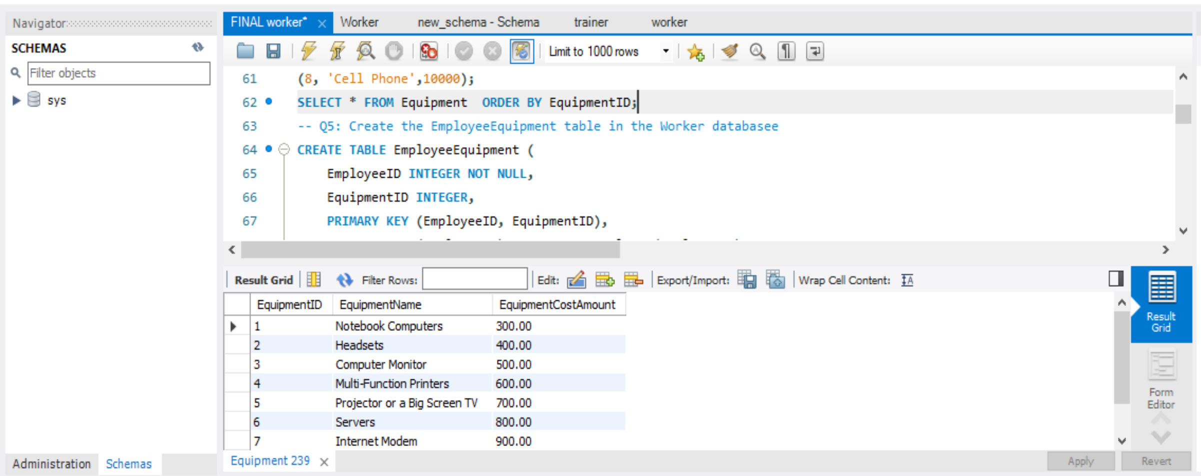


Q3. Create the Employee table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type and length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented (not in a graphical view). (c)

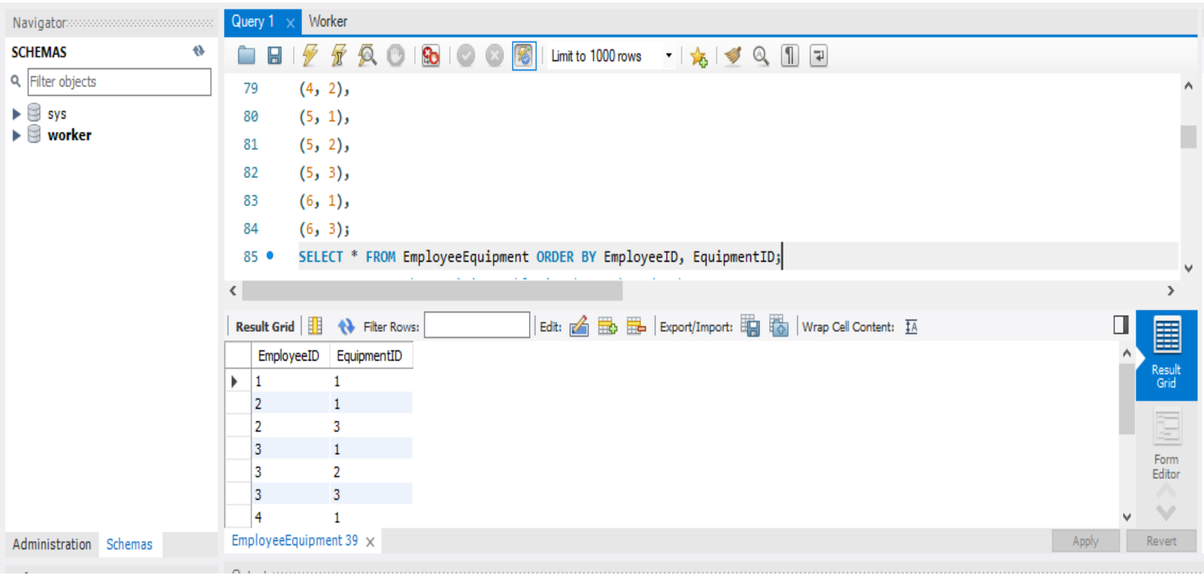
Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Employee table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



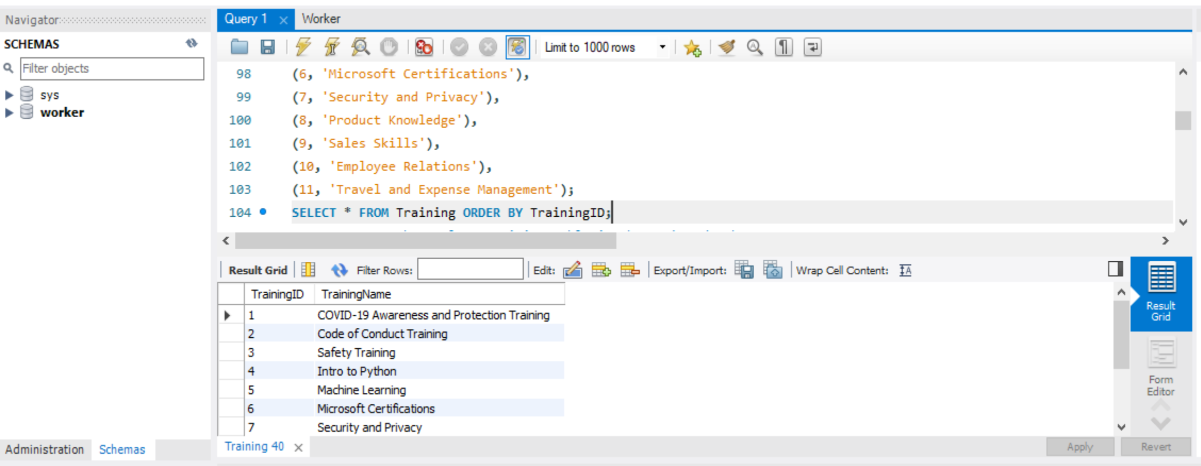
Q4. Create the Equipment table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Equipment table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



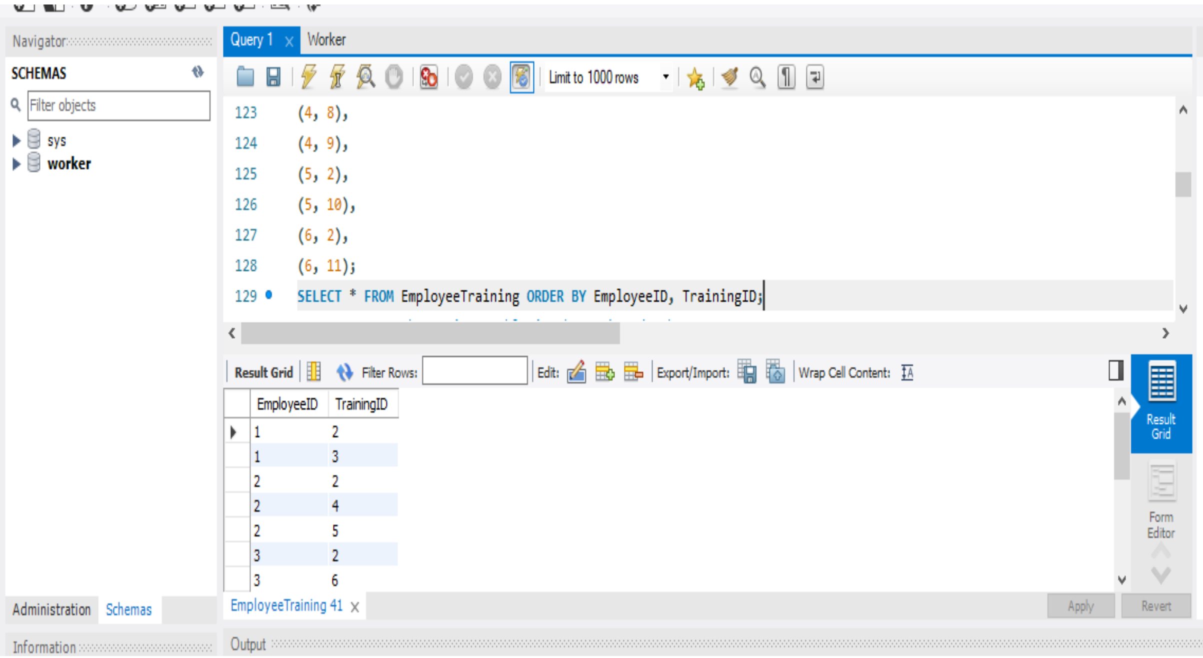
Q5. Create the EmployeeEquipment table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the EmployeeEquipment table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



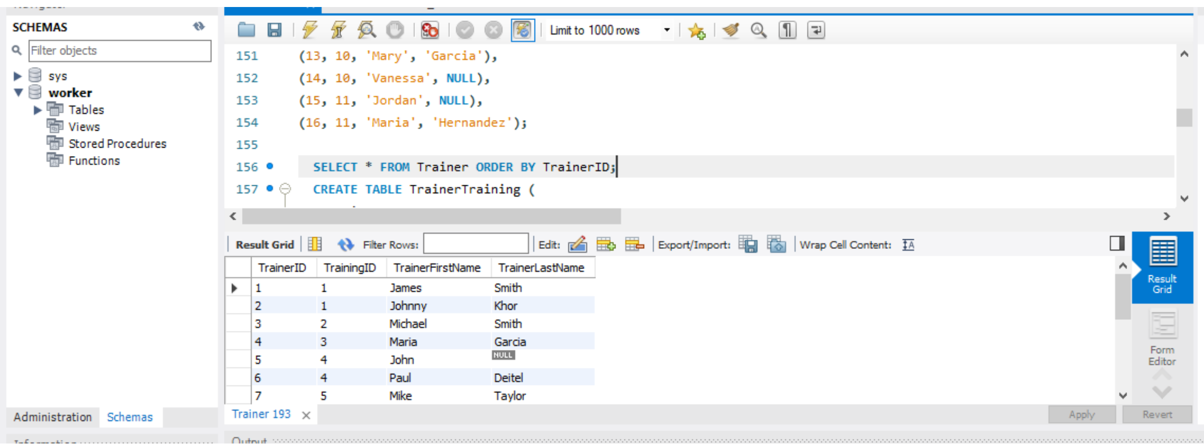
Q6. Create the Training table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Training table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



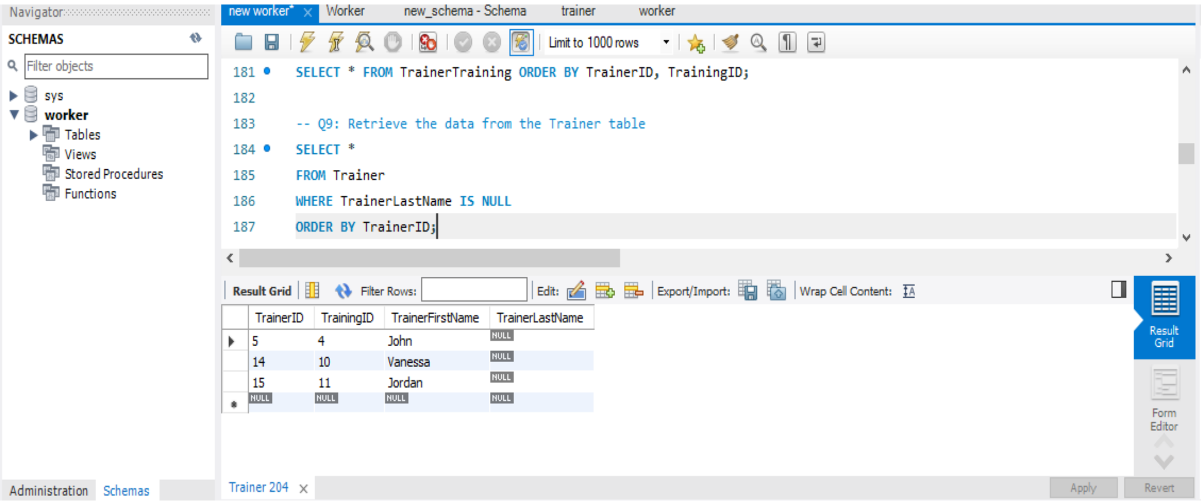
Q7. Create the EmployeeTraining table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the EmployeeTraining table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



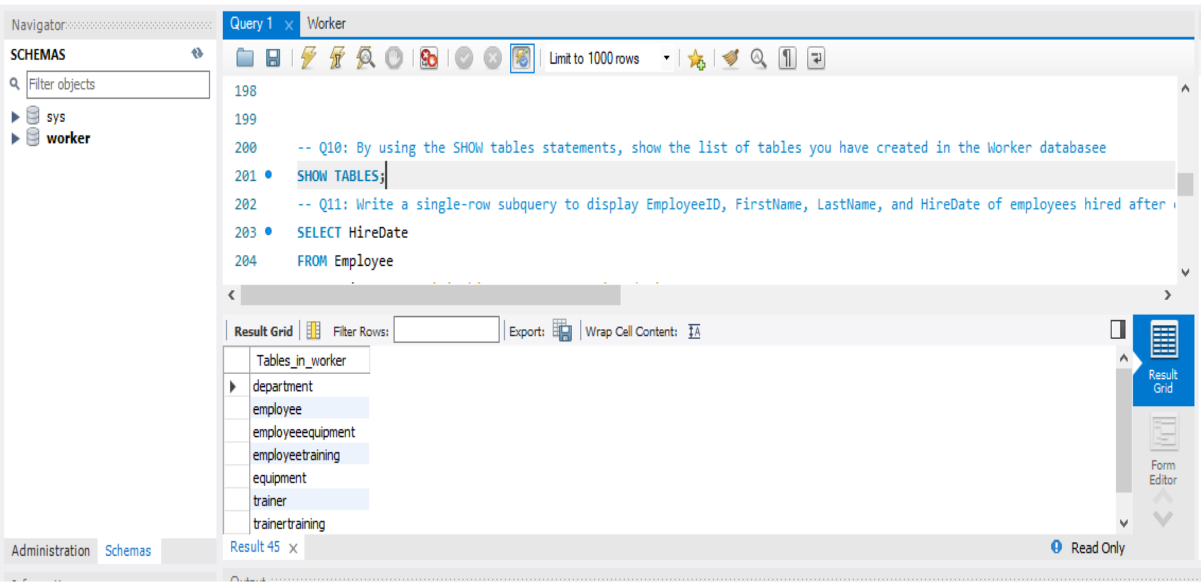
Q8. Create the Trainer table in the Worker database (table must be based on Physical Model Provided in the Assignment folder). (a) Columns, Primary Key (PK), Data Type & length, and NULL/NOT NULL need to be implemented, as provided in the Physical Model. (b) Show the table definition (DDL) that you implemented. (c) Insert the complete set of data provided in the Excel file (uploaded in the Assignment folder) and show the insert statements used. (d) Retrieve the data from the Trainer table by using the SELECT * statement and order by PK column(s). Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



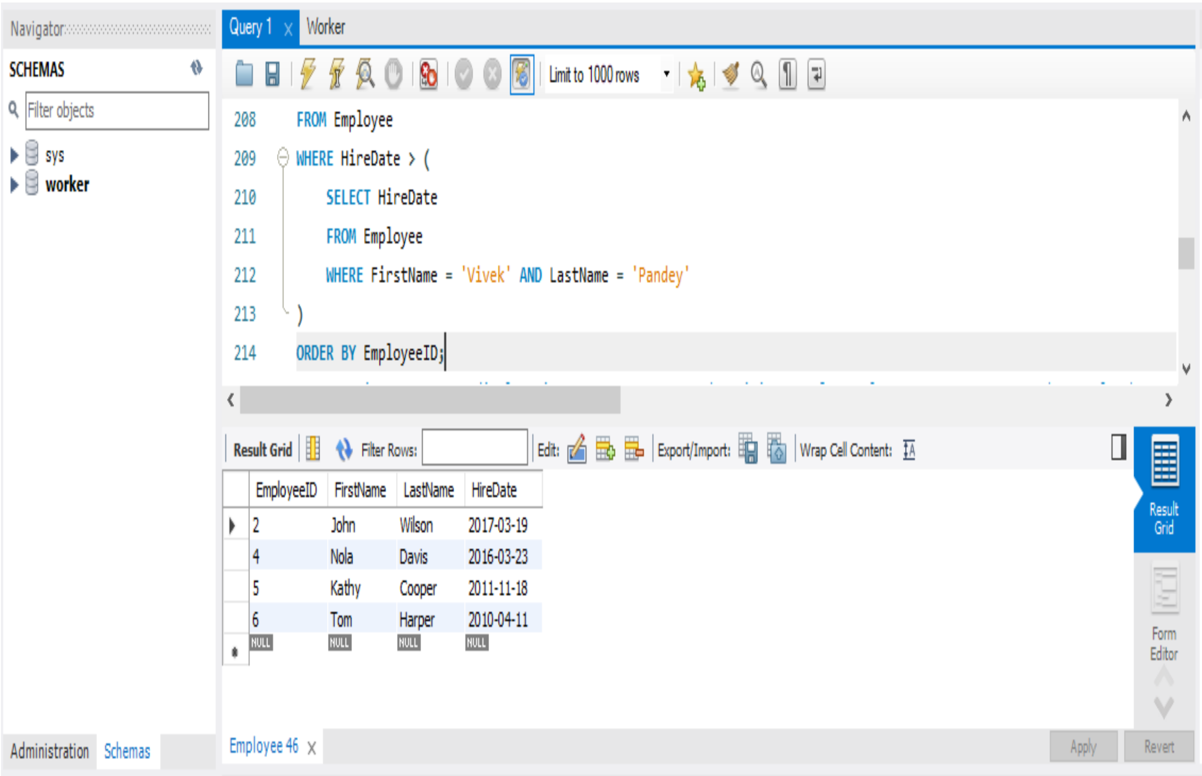
Q9. Retrieve the data from the Trainer table by using the SELECT * statement with filter, WHERE TrainerLastName IS NULL. Show the output. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



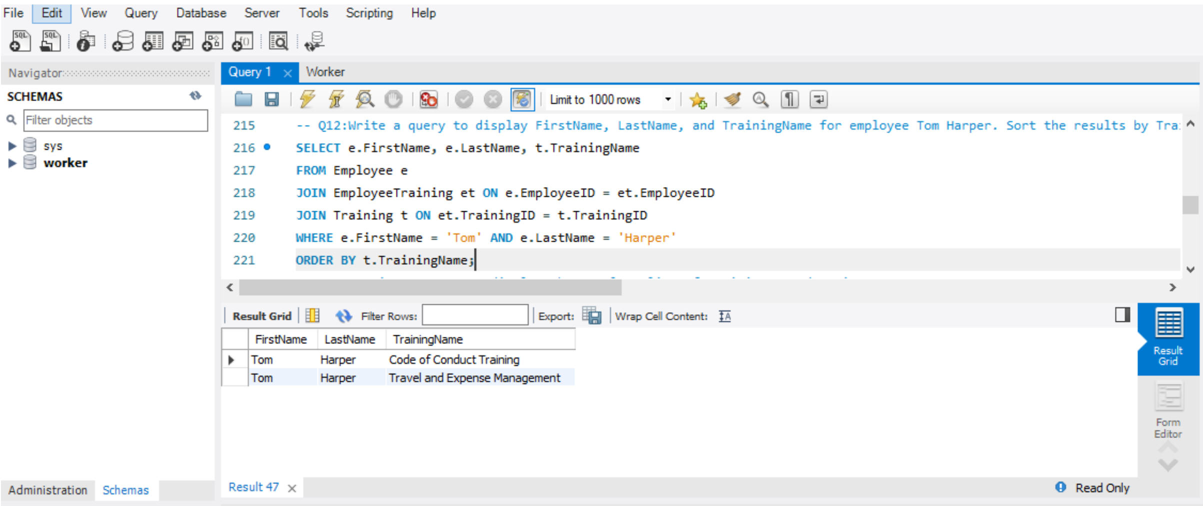
Q10. By using the SHOW tables statements, show the list of tables you have created in the Worker database. Show the screenshot of the execution of the above statements and results. Make sure you show the print screen of the complete set of the rows and columns. The rows must be ordered by PK column(s).



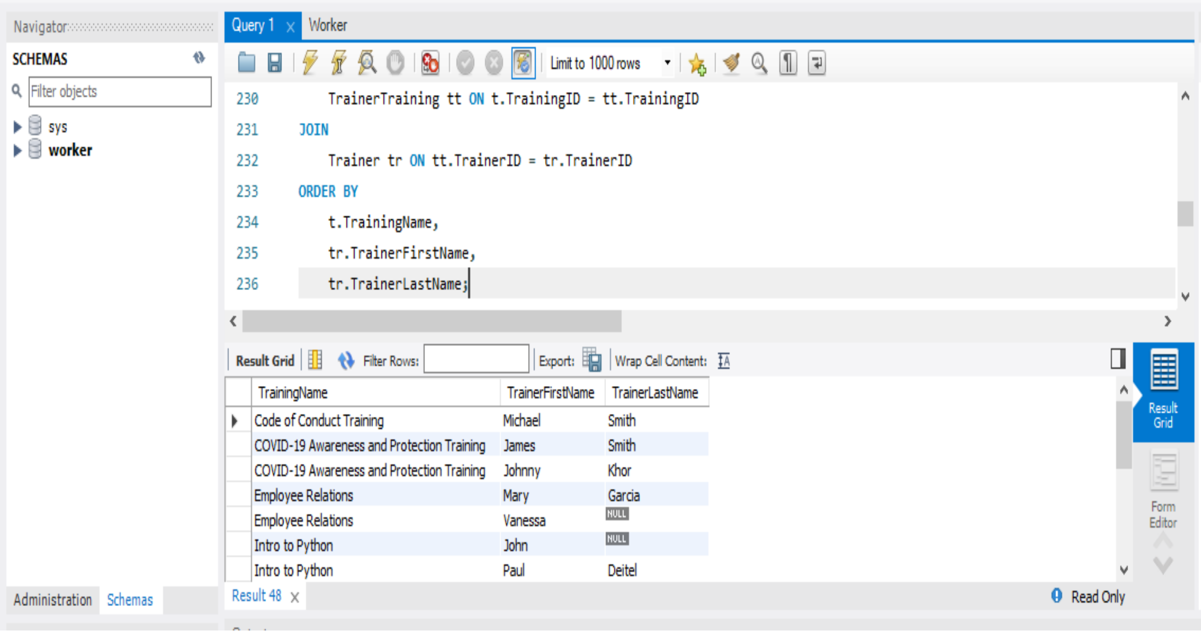
Q11. Write a single-row subquery to display EmployeeID, FirstName, LastName, and HireDate of employees hired after employee Vivek Pandey. Sort the results by EmployeeID.



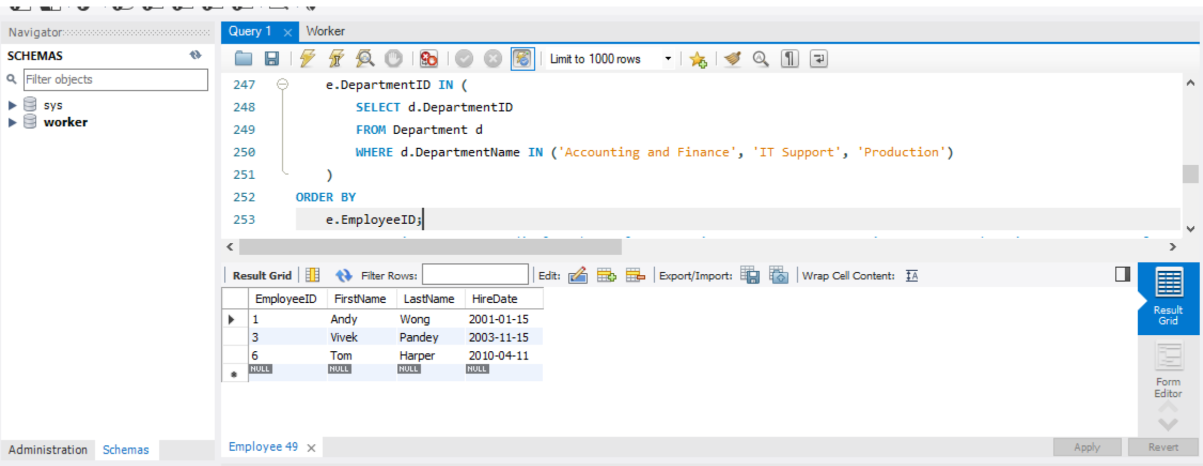
Q12. Write a query to display FirstName, LastName, and TrainingName for employee Tom Harper. Sort the results by TrainingName. Make sure you show the print screen of the complete set of the rows, and columns as specified.



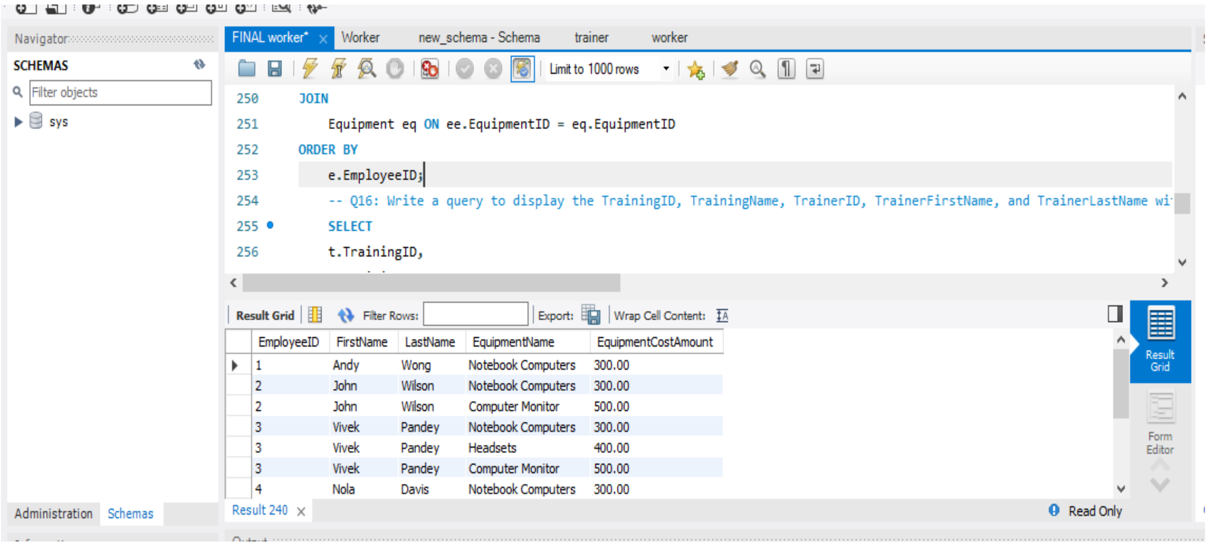
Q13. Write a query to display the complete list of Trainings, and trainers (first and last name) available for each training. Sort the output by TrainingName and Trainers' first and last name. Make sure you show the print screen of the complete set of the rows, and columns as specified.



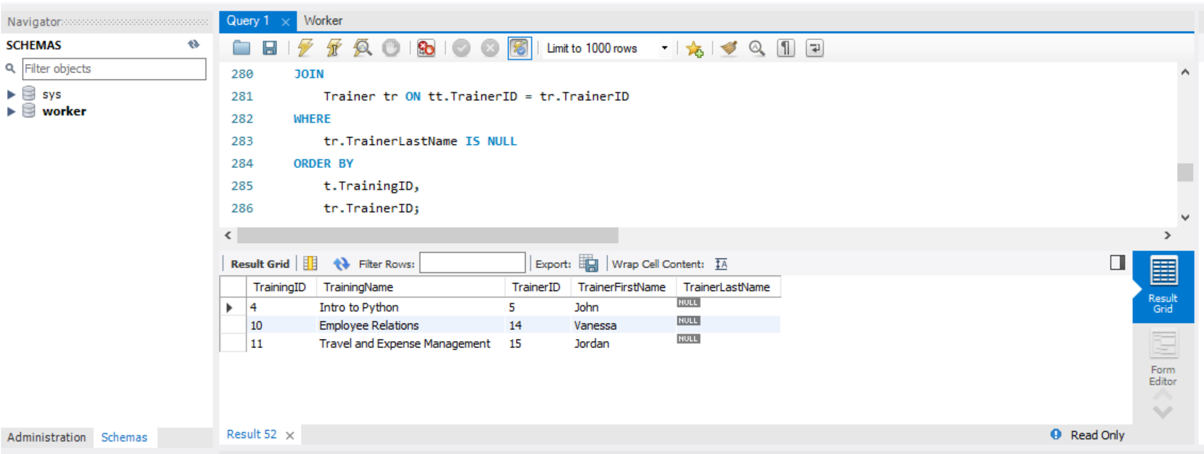
Q14. Write a multiple-row subquery to display EmployeeID, FirstName, LastName, and HireDate of employees who work for the following departments: Accounting and Finance, IT Support, and Production. Sort the results by EmployeeID. Make sure you show the print screen of the complete set of the rows, and columns as specified.



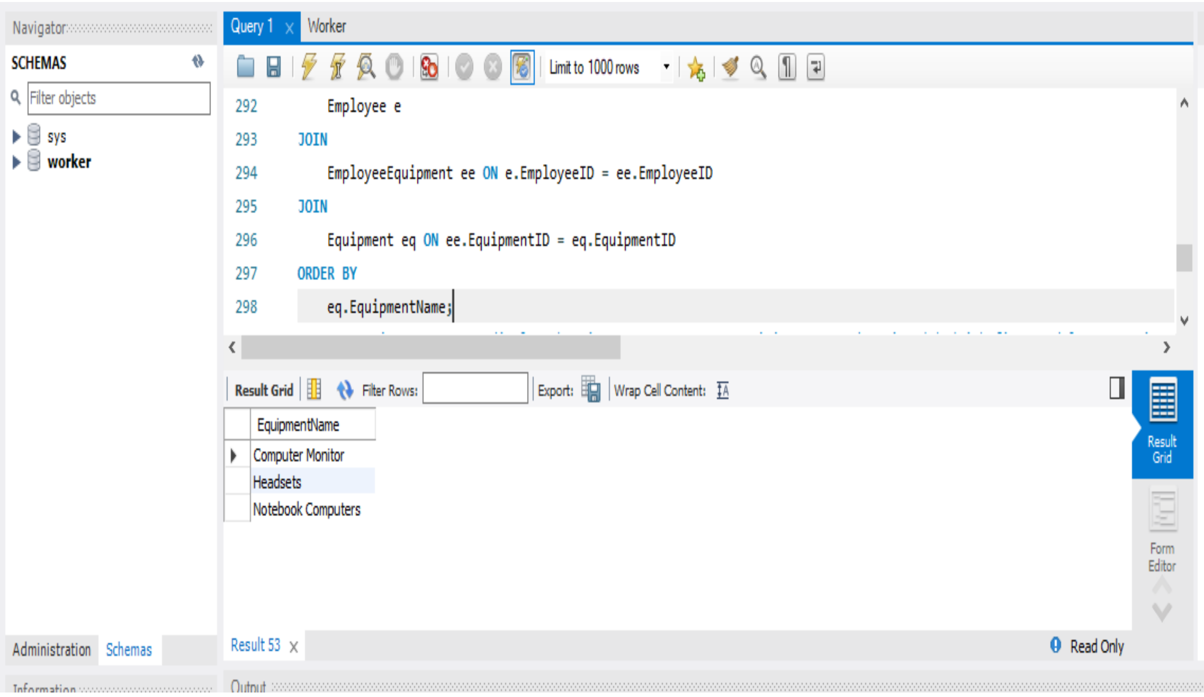
Q15. Write a query to display the EmployeeID, FirstName, LastName, EquipmentName, and EquipmentCostAmount for one of the employees. Sort the results by EmployeeID. Make sure you show the print screen of the complete set of the rows, and columns as specified.



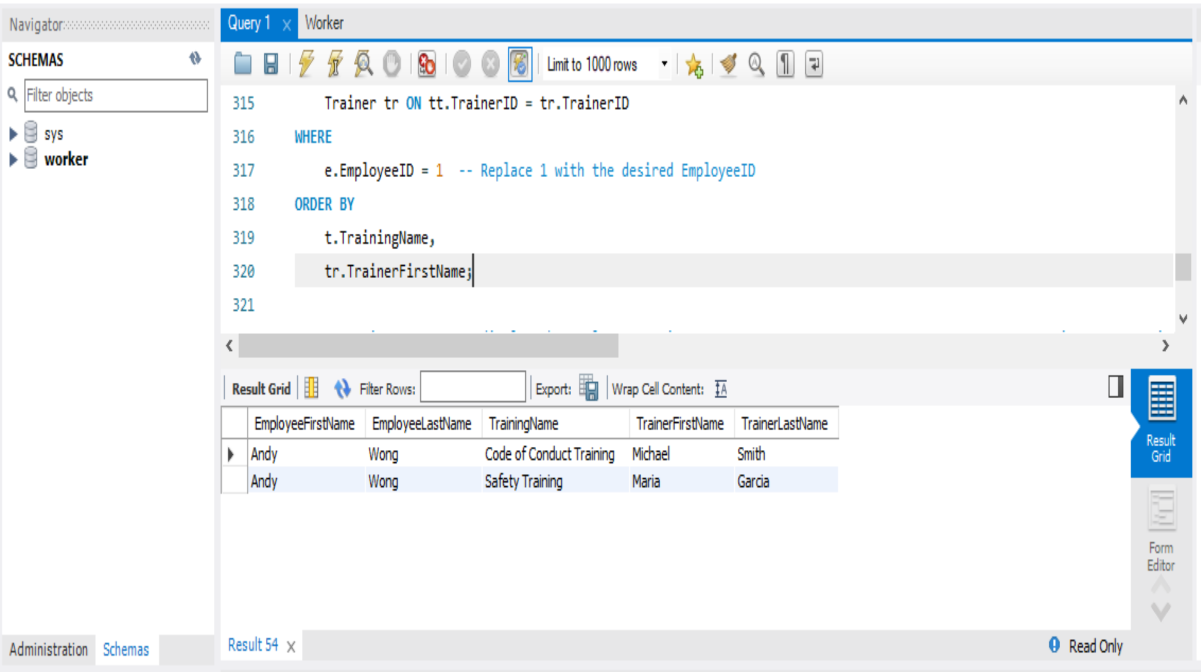
Q16. Write a query to display the TrainingID, TrainingName, TrainerID, TrainerFirstName, and TrainerLastName with the trainers who did not provide their last name. Sort the results by TrainingID and TrainerID. Make sure you show the print screen of the complete set of the rows, and columns as specified.



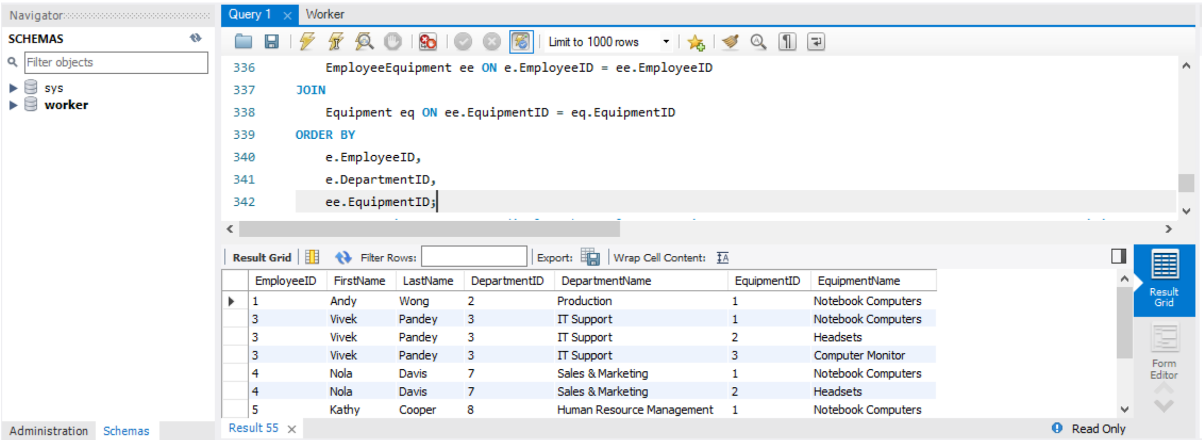
Q17. Write a query to display the distinct list of equipments used by the current employees. Sort the output by EquipmentName. Make sure you show the print screen of the complete set of the rows, and columns as specified.



Q18. Write a query to display the FirstName, LastName, TrainingName, and trainer(s) (with first and last name in two separate columns) for one of the employees. Sort the results by TrainingName and TrainerFirstName. Make sure you show the print screen of the complete set of the rows, and columns as specified.



Q19. Write a query to display the EmployeeID, FirstName, LastName, DepartmentID, DepartmentName, EquipmentID, EquipmentName for all employees. Sort the results by EmployeeID, DepartmentID, and EquipmentID. Make sure you show the print screen of the complete set of the rows, and columns as specified.



Q20. Write a query to display the EmployeeID, FirstName, LastName, DepartmentID, DepartmentName, TrainingID, TrainingName for all employees. Sort the results by EmployeeID, DepartmentID, and TrainingID. Make sure you show the print screen of the complete set of the rows, and columns as specified.